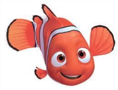
Effective two-component and two-mediator dark matter

Diego Restrepo

Instituto de Física
Universidad de Antioquia
Phenomenology Group
`http://gfif.udea.edu.co`

In collaboration with
Kimy Agudelo, Andrés Rivera and David Suárez (UdeA)



Anomaly cancellation of a $U(1)$ gauge symmetry with chiral fermions

In a fundamental theory the elementary fermions are expected to be chiral, i.e, the their charges should not allow a mass term larger than the scale of spontaneous symmetry breaking. For any set of chargues associted to a $U(1)$ gauge symmetry

$$\boldsymbol{Z} = [Z_1, Z_2, \cdots, Z_N] \ ,$$

The triangle anomaly with three $U(1)$ gauge bosons on the external lines, i.e., the $[U(1)]^3$ anomaly, and with one $U(1)$ gauge boson and two gravitons on the external lines should also be cancelled

$$\sum_{\alpha=1}^{N} Z_\alpha = 0\,, \qquad \sum_{\alpha=1}^{N} Z_\alpha^3 = 0\,, \tag{1}$$

$U(1)_Y$: no vector-like, i.e., pairs of fields with 'equal-but-opposite' charges

The hypercharge for one generation in the standard model can be seen as gauge Abelian symmetry $U(1)_Y$ with 15 left-handed Weyl massless fermions with integer charges

$$Y = [1, 1, 1, 1, 1, 1, -4, -4, -4, 2, 2, 2, -3, -3, 6]$$

which satisfy

$$\sum_i Y_i = 0\,, \qquad\qquad \sum_i Y_i^3 = 0\,.$$

If we introduce an scalar, H, of charge 3 we can form Yukawa couplings through the Dirac pairs

$$u_\alpha = (1, -4), (1, -4), (1, -4), \qquad d_\alpha = (1, 2), (1, 2), (1, 2), \qquad e = (-3, 6)$$

and the *chiral fermions* acquire Dirac masses after the EWSB. They are degenerate because the additional color symmetry. One remains massless, $\nu_L = (-3)$. The renmant Z_3 symmetry guarantees the stability of e and the other lightest quark, d_α is accidentally stable.

The two mediators of the interactions are B_μ and H.

Secluded gauge $U(1)_D$ without vector-like fermions:

$$\boldsymbol{S} = [\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

- *Dark Higgs mechanism*: Singlet dark Higgs ϕ acquires a vev and give mass to the dark photon

$$\mathcal{L} = i\psi_a^\dagger \overline{\sigma^\mu} \left(\partial_\mu - i g_D Z_\mu^D\right) \psi_a - \frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \sum_{a<b} h_{ab} \psi_a \psi_b \phi^{(*)} + \text{h.c} - V(\phi)\,. \tag{2}$$

- S_α are the charges of SM-singlet right-handed chiral fermions with $N \geq 5$
    - χ_i *massless fermions* with $i = 1, \cdots, N'$ with $N' \leq N$
    - ψ_a *multi-component dark matter*: massive after the spontaneous symmetry breaking of $U(1)_D$ with $a = N'+1, \cdots, N$
- *Larger parameter space*: Dark photon, Z_D, exclusions instead of Z'
- Z_D and ϕ masses are related by (arXiv:1610.03063)

$$h/g_D = \sqrt{2} m_\psi / m_{Z_D}\,.$$

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \cdots, \psi_{N'}]$$

$$\chi_1 \to \nu_{R1}, \cdots, \chi_{N_\nu} \to \nu_{R\,N_\nu}, \qquad 2 \leq N_\nu \leq 3\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{ij}\,(\nu_{Ri})^\dagger\,\epsilon_{ab}\,L_j^a\,H^b \left(\frac{\phi^*}{\Lambda}\right)^\delta + \text{H.c.}, \qquad \text{with } i,j = 1,2,3\,,$$

ϕ is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and give mass to *all* ψ_a. ν_R remains massless

$$\phi = -\frac{\nu}{\delta}\,,$$

Multi-component dark matter model with two-mediators: Z_D and ϕ

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[\nu, \nu, (\nu), \psi_1, \psi_2, \cdots, \psi_{N'}]$$

$$\chi_1 \to \nu_{R1}, \cdots, \chi_{N_\nu} \to \nu_{R\,N_\nu}, \qquad 2 \leq N_\nu \leq 3\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{ij} \, (\nu_{Ri})^\dagger \, \epsilon_{ab} \, L_j^a \, H^b \left(\frac{\phi^*}{\Lambda}\right)^\delta + \text{H.c.}, \qquad \text{with } i, j = 1, 2, 3\,,$$

ϕ is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and $\boxed{\text{give mass to } \textit{all } \psi_a}$. ν_R remains massless

$$\phi = -\frac{\nu}{\delta}\,,$$

Multi-component dark matter model with two-mediators: Z_D and ϕ

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \to \nu_{R1},\ \chi_2 \to \nu_{R2}, \qquad N_\nu = 2\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj}\,(\nu_{Ra})^\dagger\,\epsilon_{bc}\,L_j^b\,H^c\left(\frac{\phi^*}{\Lambda}\right) + \text{H.c.}, \qquad \text{with } j = 1, 2, 3\,,$$

ϕ is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and $\boxed{\text{give mass to } \textit{all } \psi_a}$. ν_R remains massless

$$\phi = -\nu = -5\,,$$

Two-component dark matter model with two-mediators: Z_D and ϕ

Decrease the number of charges to be assigned to dark matter particles, ψ_i below

$$[\chi_1, \chi_2, \cdots, \psi_1, \psi_2, \cdots, \psi_{N'}]$$

Secluded case:

$$[5, 5, -3, -2, 1, -6]$$

$$\chi_1 \to \nu_{R1}, \ \chi_2 \to \nu_{R2}, \qquad N_\nu = 2\,,$$

$$\mathcal{L}_{\text{eff}} = h_\nu^{aj} (\nu_{Ra})^\dagger \, \epsilon_{bc} \, L_j^b \, H^c \left(\frac{\phi^*}{\Lambda} \right) + \text{H.c.}, \qquad \text{with } j = 1, 2, 3\,,$$

ϕ is dark Higgs responsible for the SSB of the anomaly-free gauge symmetry and $\boxed{\text{give mass to } \textit{all } \psi_a}$. ν_R remains massless

$$\phi = -\nu = -5\,,$$

Two-component dark matter model with two-mediators: Z_D and ϕ

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c} - V(\phi)\,.$$

$\boxed{\text{two-component and two-mediator DM}}$ with two Dirac-fermion DM particles

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{\mathcal{D}}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c} - V(\phi)\,.$$

$\boxed{\text{two-component and two-mediator DM}}$ with two Dirac-fermion DM particles

$$\boldsymbol{z} = [5, 5, -3, -2, 1, -6] \to \phi = -5 \to [(5,5), (-3,-2), (1,-6)]$$

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume decoupled inert scalar (vev-less) sector

$$\mathcal{L} = i\psi_i^\dagger \not{D}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c} - V(\phi)\,.$$

two-component and two-mediator DM with two Dirac-fermion DM particles

$$\boldsymbol{z} = [5, 5, -3, -2, 1, -6] \to \phi = -5 \to [(5, 5), (-3, -2), (1, -6)]$$

$$\mathcal{L} \subset h_{(-3,-2)}\psi_{-3}\psi_{-2}\phi^* + h_{(1,-6)}\psi_1\psi_{-6}\phi^* + \text{h.c.}$$

Minimal secluded model with D-5 effective Dirac neutrino masses

We assume *a minimal* decoupled inert scalar (vev-less) sector

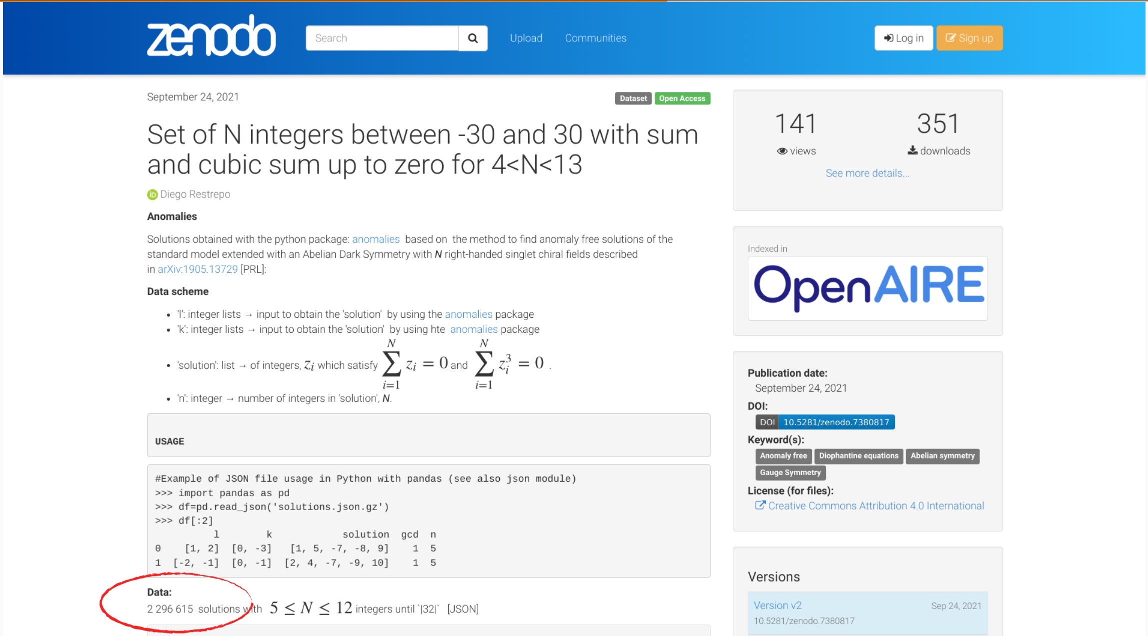
$$\mathcal{L} = i\psi_i^\dagger \not{D}\psi_i - \frac{1}{4}V_{\mu\nu}V^{\mu\nu} + \sum_{i<j} h_{ij}\psi_i\psi_j\phi^{(*)} + \text{h.c} - V(\phi)\,.$$

two-component, two-mediator DM with two-flavor scotogenic sector

$$\boldsymbol{z} = [-9,-9,-9,10,10,-1,-1,4,5] \to \phi = 9 \to [(-9,-9,-9),(10,-1),(10,-1),(4,5)]$$

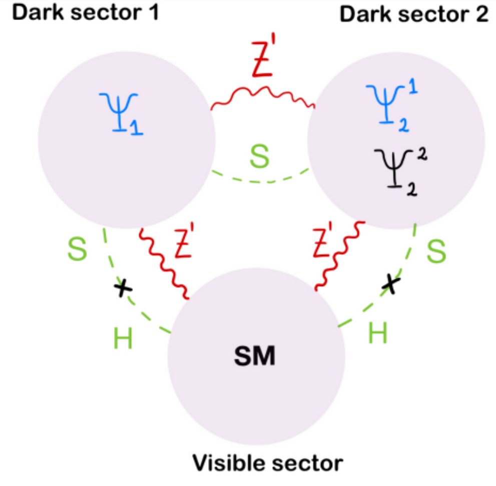
$$\mathcal{L} \supset \underbrace{\sum_{a,b=1}^{2} h_{(10a,-1b)}\psi_{10}\psi_{-1}\phi}_{\text{multi-flavor scotogenic DM}} + h_{(4,5)}\psi_4\psi_5\phi^* + \text{h.c.} \tag{3}$$

https://doi.org/10.5281/zenodo.7380817



Solutions ~ 1000	N	ν	δ	ϕ	N_D	N_M	G_D	G_M
$(5,5,-2,-3,1,-6)$	6	5	1	-5	2	0	1	0
$(-9,-9,-9,10,10,-1,-1,4,5)$	9	9	1	-9	3	0	2	0
$(1,-2,3,4,6,-7,-7,-7,9)$	9	-7	1	7	3	0	1	0
$(1,-2,-2,3,3,-4,-4,6,6,-7)$	10	6	1	**-6**	3	2	2	2
$(1,-2,-2,3,4,-5,-5,7,7,-8)$	10	-5	1	5	4	0	2	0
$(1,-2,-2,3,5,-6,-6,8,8,-9)$	10	-6	1	**6**	4	0	2	0
$(2,2,3,4,4,-5,-6,-6,-7,9)$	10	2	1	-2	4	2	2	2
$(1,1,5,5,5,-6,-6,-6,-9,10)$	10	1	1	-1	4	0	3	0
$(2,2,4,4,-7,-7,-9,-9,10,10)$	10	10	2	-5	3	0	2	0
$(1,2,2,-3,6,6,-8,-8,-9,11)$	10	-8	1	8	4	1	2	1
$(1,-2,-3,5,6,-8,-9,11,11,-12)$	10	11	1	-11	4	0	1	0
$(1,1,-3,4,4,-7,8,-10,-10,12)$	10	-10	2	5	4	0	2	0
$(1,1,-2,-2,-4,6,-10,11,12,-13)$	10	-2	1	2	3	2	1	2
$(3,4,4,4,4,-5,-8,-8,-11,13)$	10	-8	1	8	2	4	1	4
$(4,4,5,6,6,-9,-10,-10,-11,15)$	10	6	1	**-6**	4	0	2	0
$(1,-2,-4,7,7,-10,-12,14,14,-15)$	10	14	1	**-14**	3	2	1	2
$(1,2,2,-3,4,-6,12,-13,-14,15)$	10	2	1	-2	4	1	1	1
$(1,4,4,-7,8,8,-9,-12,-12,15)$	10	8	1	-8	4	2	2	2
$(1,2,2,-9,-9,16,16,17,-18,-18)$	10	-18	1	18	3	2	2	2
$(1,-3,-6,7,-10,11,-16,18,18,-20)$	10	18	2	-9	4	0	1	0
$(1,-4,5,-6,-6,10,-14,15,20,-21)$	10	-6	1	**6**	4	0	1	0
$(2,-3,-6,7,12,-14,-14,17,20,-21)$	10	-14	1	**14**	4	1	1	1
$(3,6,6,-7,8,8,-14,-14,-17,21)$	10	-14	1	**14**	4	1	2	1
$(8,8,9,10,10,-13,-18,-18,-27,31)$	10	-18	1	18	4	1	2	1

Effective two-component and two-mediator DM without kinetic mixing: $Z_D \to \overline{\nu}\nu$



Decoupled inert scalar sector $\to \Delta N_{\mathrm{eff}} = 0$

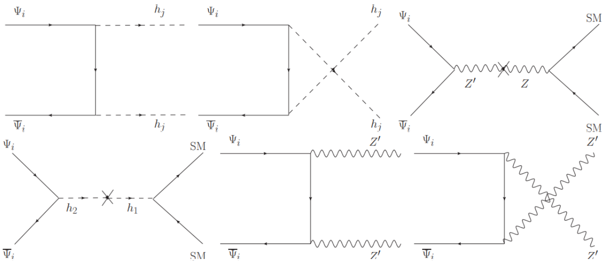


Figure 3: DM annihilation channels.

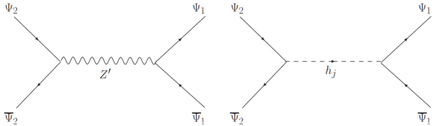
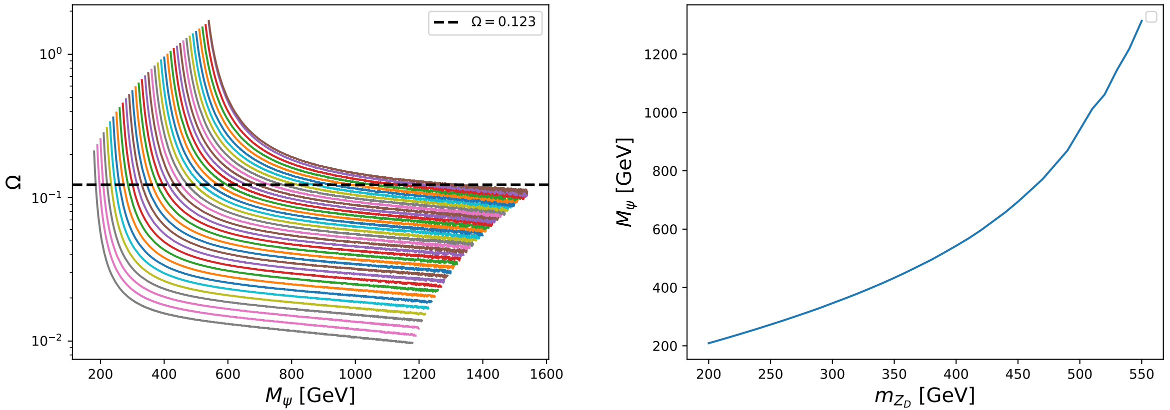


Figure 4: DM conversion channels $\Psi_1 \leftrightarrow \Psi_2^i$.

Decoupled inert scalar sector $\to \Delta N_{\text{eff}} = 0$

$M_2 \gg M_1$ and $g_D = 0.1$



Conclusions